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Evaluation of bioactivities and phenolic contents of wild edible mushrooms from northeastern Thailand

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Abstract Twenty-five wild edible mushrooms from Northeastern Thailand were analyzed for their antioxidant activities, proteins, sugars, β-glucan, and phenolic profiles. Results showed that *T. clypeatus* and *V. volvacea* exhibited the greatest scavenging activity (83.07 and 86.60%) and reductive power (9.79 and 8.42 g Fe(II)/kg, respectively). *T. clypeatus* recorded the highest content of (+)-catechin and naringenin (13.40 and 0.74 g/kg dw), with *V. volvacea* the highest amount of quercetin and quercetin-3-*O*-rutinoside (1.82 and 1.16 g/kg dw, respectively). Both *T. clypeatus* and *V. volvacea* also exhibited the greatest amounts of β-glucan (125.23 and 344.43 g/kg dw) and protein (343.30 and 452.20 g/kg dw, respectively) among the mushroom species evaluated. Results suggested that both *T. clypeatus* and *V. volvacea* were a good source of healthy compounds, namely β-glucan and flavonoids, and could be used to mitigate diseases involving free radicals.

Keywords Phenolics · Flavonoids · Antioxidants · Mushrooms · Reductive power

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Introduction

Wild edible mushrooms have long been exploited by many Asian countries as food and medicine [1] and are now becoming more important in the human diet with their nutritional and culinary values [2]. Mushrooms have high protein, low fat and cholesterol contents, and are considered by many as an ideal source of nutrition, rich in minerals and vitamins [3]. Edible mushrooms accumulate a variety of secondary metabolites in their fruiting bodies including phenolic compounds, polysaccharides, polyketides, terpenes and steroids. Phenolic compounds are an excellent source of antioxidants and possess anti-tumor, cardiovascular, anti-viral, anti-microbial, anti-inflammatory, and anti-allergenic properties. They also help to balance blood sugar levels and support the body's detoxification mechanisms [4–6]. Furthermore, numerous companies are developing capsules from mushroom combinations, and although expensive, these have proved beneficial to health by counteracting cancer [7].

Northeastern Thailand has the highest wild edible mushroom diversity in Asia, and some mushrooms have high gastronomic relevance and/or medicinal properties. The Phu Phan National Park, located in Northeastern Thailand, is well known for its diversified natural resources which offers favorable environmental conditions for the growth of wild edible mushrooms. In addition to the health benefits, these mushrooms are a food source with potential economic value for the local people. A recent study by Seephonkai et al. [8], reported that mushrooms in the genus *Phellinus* collected from Northeast Thailand exhibited excellent antioxidant activity.

In view of the mushroom species biodiversity in Northeast Thailand and their interesting pharmacological activities, the antioxidant properties and phenolic

Table 1 Natural habitats of twenty-five wild edible mushroom species studied for bioactive compound content and antioxidant potential

Local name	Scientific name	Family	Natural habitat
Gilled mushroom			
Het Ra Ngok Kao	*Amanita princeps*	Amanitaceae	Saprophytic (grows on the ground under the shade of trees, *Dipterocarpus alatus*, *Hopea odorata* and *Anisoptera costata*)
Het Ra Ngok Luang	*Amanita hamibapa*	Amanitaceae	Saprophytic (on the ground under the shade of trees, *Dipterocarpus alatus*, *Hopea odorata* and *Anisoptera costata*)
Het Hat	*Lactarius volemus*	Russulaceae	Saprophytic (grows on the ground in open forests)
Het Nam Mag	*Russula luteotacta*	Russulaceae	Saprophytic (grows on the ground in open forests)
Het Nam Mag Yai	*Russula emetica*	Russulaceae	Saprophytic (grows on the ground in open forests)
Het Na Kao	*Russula alboareolata*	Russulaceae	Saprophytic (grows on the ground in open forests)
Het Kow Kao	*Russula galochroides*	Russulaceae	Saprophytic (grows on the ground under *Lithocarpus cantleyanus*)
Het Kow	*Russula cyanoxantha*	Russulaceae	Saprophytic (grows on the ground under *Lithocarpus cantleyanus*)
Het Tan Yai	*Russula nigricans*	Russulaceae	Saprophytic (grows on the ground under *Dipterocarpus alatus*)
Het Tan Noi	*Russula densfolia*	Russulaceae	Saprophytic (grows on the ground under *Dipterocarpus alatus*)
Het Kai	*Russula delica*	Russulaceae	Saprophytic (grows on the ground under *Shorea obtuse* and *Shorea siamensis*)
Het Na Lae	*Russula violeipes*	Russulaceae	Saprophytic (grows on the ground in open forests)
Het Pluak Tab	*Termitomyces fuliginosus*	Amanitaceae	Symbiotic (in association with termite nests)
Het Pluak Jig	*Termitomyces clypeatus*	Amnitacca	Symbiotic (in association with termite nests)
Het Teen Rat	*Tricholoma crassum*	Tricholomataceae	Saprophytic (grows naturally in various habitats such as meadows, rice fields, forests and soil containing high organic matter)
Het Fang	*Volvariella volvacea*	Pluteaceae	Saprophytic (grown on rice straw)
Puffball mushroom			
Het Pow Nung	*Astraeus hygrometricus*	Astraeaceae	Saprophytic (grows on the ground under the shade of trees, *Dipterocarpus alatus*)
Het Klum Pra	*Alpova trappei*	Sclerodertraceae	Saprophytic (grows on the ground in open forests)
Shelf mushroom			
Het Hoo Noo	*Auricularia auricula*	Auriculariaceae	Saprophytic (grows on both dead and living wood)
Het Man Poo	*Cantharellus cibarius*	Cantharellaceae	Saprophytic (grows on the ground in woods especially *Mangifera indica*)
Hed Ma Het Man Poo Yai	*Cra Craterellus aureus*	Cantharellaceae	Saprophytic (grows on the ground in woods especially *Mangifera indica*)
Het Kon Kao	*Lentinus squarrosulas*	Polyporaceae	Saprophytic (grows on decaying plants of *Shorea obtuse*, *Shorea siamensis* and *Mangifera indica*)
Het Bod	*Lentinus polychrous*	Polyporaceae	Saprophytic (grows on decaying plants of *Hopea odorata* and *Anisoptera costata*)
Het Hom	*Lentinus edodes*	Tricholomataceae	Saprophytic (grows on decaying plants of *Lithocarpus cantleyanus*)
Het Tong Fon	*Lentinus giganteus*	Polyporaceae	Saprophytic (grows on the ground in woods especially *Mangifera indica*)

constituents of twenty-five species of wild edible mushrooms were investigated. This is the first study on the phenolic composition and antioxidative activities of wild edible mushrooms collected from native forests in Northeastern Thailand.

Materials and methods

Reagents and chemicals

Deionized water was prepared by a Milli-Q Water Purification system (Millipore, MA, USA). Acetonitrile, methanol, and phosphoric acid were of HPLC grade from Merck (Darmstadt, Germany). Folin-Ciocalteu reagent and TPTZ (2,4,6-tripyridyl-*s*-triazine) were purchased from Sigma-Aldrich Chemical Company (St. Louis, MO, USA). DPPH (2,2-diphenyl-1-picryl hydrazyl hydrate) was obtained from Fluka (Buchs, Switzerland). Pure standard phenolics, including myricetin, quercetin, rutin, (+)-catechin, (−)-epicatechin, naringenin and kaempferol were purchased from Sigma-Aldrich Chemical Company (St. Louis, MO, USA). β-1,3-glucan was obtained from Sigma-Aldrich (Seelze, Germany).

Mushroom sample preparation

Twenty-five species of mushroom fruiting bodies were purchased from the indigenous people who collect from forest areas of the Phu Phan Mountains within Phu Phan National Park, Sakon Nakhon Province between June and August 2016. The area is 400–567 m above sea level. The mushrooms were identified by Dr. Sujitar Jorjong, Department of Plant Science, Faculty of Natural Resources, Rajamangala University, Thailand. The names of the species studied and their natural habitats are presented in Table 1. The mushrooms were classified into three groups as gilled mushrooms, puffball mushrooms and shelf mushrooms.

Two hundred and fifty samples of the twenty-five wild edible species collected (10 subsamples from each species) were analyzed for their phenolic composition and antioxidant activity. The mushroom samples were washed, air dried, and then dried in an oven at 60 °C. The dried mushrooms were ground to a fine powder (ca. 1 mm size) and stored in airtight plastic bags in desiccators at room temperature prior to analysis. A 1 g sample of each dried mushroom was mixed with 10 ml of 60% methanol and subjected to extraction by an ultrasonic cleaner (KUDOS Ultrasonic Equipment Co., Shanghai, China) for 1 h at room temperature. The samples were then centrifuged at 2,000 *g* for 15 min, the solvent was decanted and the residue was re-extracted with 10 ml of fresh the solvent. The two extracts were combined and the volume was adjusted to 20 ml. Appropriate aliquots of extracts were filtered through a 0.45 μm membrane filter and stored at −18 °C until required for analysis.

Protein analysis

Total nitrogen concentration was measured by elemental analysis (LECO TruSpec Micro CHNS Determinator, LECO Corporation, St. Joseph, MI, USA). Samples (2 mg) were placed in tin capsules in an oven for combustion at 1100 °C using pure oxygen (20 cm^3) as the combustion gas and pure helium as the carrier gas. Carbon, hydrogen and sulfur were determined by infrared absorption while nitrogen was measured as N_2 using a thermal conductivity detection system.

HPLC analysis for β-glucan, sugars, and phenolic compounds

The concentrations of β-glucan and sugars were determined using HPLC according to Kelebek et al. [9] (Shimadzu Cooperation Analytical & Measuring Instruments Division Kyoto, Japan liquid chromatograph equipped with LC-20AD series pumping system, SIL-20A series auto injector system, RID-10A series refractive index detector, and Bio-Rad (Richmond, CA) Aminex HPX-87P (lead-based), 0.78 × 30 cm, 5 μm chromatograph column). Twenty microliters samples were injected into the column and separation was conducted at 85 °C with the mobile phase being deionized water at a flow rate of 0.6 ml/min. The identification of β-glucan and sugars in the samples was achieved by comparing retention times and spectra with those of authentic standards, and quantified by the external standard method.

HPLC was also used for determining the concentrations of specific phenolic compounds. The liquid chromatographic apparatus (Shimadzu LC-20AD) consisted of a pump (Shimadzu LC-20AD), an automatic injector (Shimadzu SIL-10AD) (20 μl injection volume) and controller coupled to a photodiode array detector (SPD-M20A) (set at 254 nm) interfaced to LC Solution software (Shimadzu Cooperation Analytical & Measuring Instruments Division Kyoto, Japan)). Separations were performed on an Apollo C_{18} column (Alltech Associates, Deerfield, IL, USA) (4.6 mm × 250 mm, 5 μm) preceded by separation on an Inertsil ODS-3 guard column (4.0 mm × 10 mm, 5 μm; GL Science Inc., Tokyo, Japan). Both columns were placed in a column oven set at 40°C. Solvent A was acetonitrile in deionized water (2% v/v), acidified with phosphoric acid (0.2% v/v), while solvent B was acetonitrile in deionized water (97.8% v/v), acidified with phosphoric acid (0.2% v/v). The elution profile started with 20% solvent B, 50%

Table 2 β-glucan, protein and sugar concentration in twenty-five species of wild edible mushrooms in Thailand

Mushroom	β-Glucan (g kg^{-1} dw)	Protein (g kg^{-1} dw)	Sugar (g kg^{-1} dw)			Sugar alcohol (g kg^{-1} dw)				
			Sucrose	Glucose	Fructose	Myo-inositol	Mannitol	Arabitol	Xylitol	Sorbitol
Gilled mushrooms										
A. princeps	90.10 ± 5.31	372.60 ± 12.50	50.48 ± 1.83	7.23 ± 0.68	4.35 ± 0.60	ND	12.99 ± 1.61	9.34 ± 1.27	0.95 ± 0.02	0.84 ± 0.07
A. hamibapa	84.92 ± 6.51	359.81 ± 10.46	83.11 ± 5.23	21.57 ± 1.57	13.19 ± 0.63	ND	7.18 ± 0.53	1.18 ± 0.80	0.88 ± 0.03	ND
L. volemus	80.39 ± 6.13	289.24 ± 17.19	0.15 ± 0.02	10.34 ± 0.22	19.63 ± 1.21	ND	164.58 ± 13.16	ND	ND	ND
R. luteotacta	121.74 ± 5.49	306.37 ± 24.50	2.77 ± 0.07	ND	6.08 ± 0.11	ND	126.13 ± 7.53	50.44 ± 5.52	2.66 ± 0.20	1.32 ± 0.10
R. emetica	36.90 ± 6.62	301.23 ± 13.16	20.48 ± 1.02	1.72 ± 0.07	21.56 ± 5.02	4.49 ± 0.28	107.36 ± 12.35	ND	ND	ND
R. alboareolata	62.69 ± 5.49	320.84 ± 10.23	0.56 ± 0.01	ND	18.39 ± 0.43	ND	116.28 ± 1.25	ND	ND	ND
R. galochroides	81.60 ± 9.55	222.30 ± 18.75	2.73 ± 0.06	ND	9.11 ± 0.35	ND	132.87 ± 9.40	53.75 ± 5.18	0.90 ± 0.04	1.14 ± 0.03
R. cyanoxantha	59.57 ± 0.53	349.06 ± 15.14	8.48 ± 0.62	2.40 ± 0.14	13.61 ± 0.70	5.99 ± 0.14	131.42 ± 13.44	ND	0.72 ± 0.03	ND
R. nigricans	76.93 ± 6.69	285.51 ± 13.22	4.58 ± 0.18	23.86 ± 7.04	1.52 ± 0.02	ND	98.03 ± 7.78	38.12 ± 4.64	0.89 ± 0.10	1.06 ± 0.04
R. densfolia	88.18 ± 5.09	254.72 ± 19.21	1.67 ± 0.04	7.71 ± 0.05	43.23 ± 2.95	ND	115.26 ± 1.30	ND	ND	ND
R. delica	86.67 ± 8.44	348.05 ± 16.17	0.92 ± 0.10	0.18 ± 0.01	2.54 ± 0.18	ND	106.32 ± 8.78	39.92 ± 5.11	3.05 ± 0.16	1.60 ± 0.08
R. violeipes	136.61 ± 7.93	397.32 ± 21.20	6.35 ± 0.14	0.07 ± 0.01	17.68 ± 1.82	0.53 ± 0.10	43.52 ± 4.61	ND	ND	ND
T. fuliginosus	115.21 ± 5.10	444.31 ± 20.35	54.29 ± 3.41	0.23 ± 0.03	5.83 ± 0.16	ND	2.56 ± 0.14	ND	1.05 ± 0.02	1.73 ± 0.03
T. clypeatus	125.23 ± 4.22	343.37 ± 15.33	31.63 ± 2.10	0.06 ± 0.01	8.56 ± 0.20	ND	10.24 ± 1.10	8.20 ± 0.16	3.54 ± 0.11	1.39 ± 0.05
T. crassum	98.26 ± 8.72	487.10 ± 17.41	90.27 ± 7.53	8.19 ± 0.40	6.60 ± 023	ND	1.26 ± 0.12	1.76 ± 0.19	0.95 ± 0.05	11.34 ± 0.05
V. volvacea	344.39 ± 11.33	452.23 ± 25.67	27.15 ± 2.10	5.77 ± 0.30	9.68 ± 0.46	ND	10.66 ± 1.19	ND	2.48 ± 0.30	3.28 ± 0.10
Puffball mushrooms										
A. hygrometricus	30.07 ± 7.24	264.81 ± 10.39	2.14 ± 0.09	2.40 ± 0.14	4.12 ± 0.20	ND	9.66 ± 0.89	2.17 ± 0.14	ND	ND
A. trappei	6.17 ± 1.13	241.34 ± 12.43	1.14 ± 0.05	ND	24.05 ± 1.63	ND	ND	ND	ND	ND
Shelf mushrooms										
A. auricula	ND	73.42 ± 6.12	ND	ND	ND	ND	ND	ND	ND	ND
C. cibarius	132.91 ± 4.27	193.91 ± 11.28	1.41 ± 0.07	ND	9.89 ± 0.53	ND	0.87 ± 0.05	ND	5.51 ± 0.15	ND
Cra C. aureus	23.13 ± 3.41	223.84 ± 17.18	12.70 ± 1.33	ND	23.85 ± 7.41	ND	37.91 ± 1.28	ND	ND	ND
L. squarrosulas	191.85 ± 8.25	404.31 ± 20.37	8.54 ± 0.36	0.64 ± 0.04	20.71 ± 0.50	ND	6.99 ± 0.33	0.67 ± 0.01	2.88 ± 0.02	1.15 ± 0.04
L. polychrous	102.93 ± 6.07	355.15 ± 24.50	31.95 ± 2.79	7.49 ± 0.68	13.23 ± 0.68	2.33 ± 0.05	13.41 ± 1.71	2.97 ± 0.30	1.39 ± 0.03	2.58 ± 0.01
L. edodes	119.31 ± 7.75	376.53 ± 16.44	155.61 ± 16.37	5.60 ± 0.18	57.11 ± 2.54	ND	118.20 ± 6.33	ND	ND	ND
L. giganteus	88.40 ± 3.13	322.17 ± 13.15	3.75 ± 0.08	9.19 ± 1.64	18.35 ± 1.81	ND	8.16 ± 0.41	ND	ND	ND

Results are expressed as mean values ± SD (n = 10). *ND* not detected

solvent B at 30 min, 60% solvent B at 35 min, 20% solvent B at 40 min and re-equilibration of the column for 15 min at a flow rate of 0.6 ml/min. Components were identified by comparing their retention times and spectra with those of authentic standards and quantified by the external standard method.

Determination of total phenolic content (TPC)

The TPC in the mushroom samples was estimated using a Folin-Ciocalteu assay, following the method of Singleton and Rossi [10] with some modifications. Briefly, 12.5 µl of the sample was mixed with 12.5 µl of Folin-Ciocalteu reagent (1:9; Folin-Ciocalteu reagent: distilled water) and 12.5 µl of distilled water. After 6 min, 125 µl of 7% sodium carbonate (Na_2CO_3) solution and 100 µl of distilled water were added to the mixture and allowed to react for 90 min. Absorbance was then read at 760 nm using a microplate reader (Synergy HT, BiotTek Instruments, USA). Gallic acid at different concentrations was used as a standard. Samples were measured in three replicates.

Determination of total flavonoid content (TFC)

The aluminum chloride ($AlCl_3$) method was used to determine the TFC of the sample extracts according to Kim et al. [11] with some modifications. Briefly, 25 µl of the diluted sample or (+)-catechin was mixed with 125 µl of deionized distilled water (ddH_2O) and 7.5 µl of 5% sodium nitrite ($NaNO_2$) solution. After 5 min, incubation at room temperature, 15 µl of 10% $AlCl_3$ solution was added. After 5 min 50 µl of 1 M sodium hydroxide (NaOH) solution was added. The mixture was immediately diluted by the addition of 27.5 µl of ddH_2O, thoroughly mixed, and its absorbance read at 510 nm against a blank prepared with ddH_2O, using a microplate reader. (+)-Catechin at different concentrations was used as a standard. Samples were measured in three replicates. The results were expressed as g (+)-catechin equivalent per kg dry weight (g CE/kg dw).

DPPH radical scavenging activity

The scavenging activity of the mushroom samples was estimated according to the procedure described by Brand-Williams et al. [12] with some modifications. Firstly, 100 µl of the sample (1 g/l) was added to 100 µl of 0.2 mM DPPH methanolic solution. The mixture was mixed thoroughly, kept in the dark at room temperature for 1 h and then absorbance was measured at 520 nm using a microplate reader. Extraction solvent was used as a blank while the control was prepared by mixing 100 µl of DPPH and 100 µl of methanol. The scavenging effect was calculated based on the following equation:

$$\text{DPPH scavenging effect } (\%) = ((A_0 - A_1)/A_0) \times 100,$$

where A_0 is the absorbance of the control and A_1 is the absorbance in the presence of the sample.

Ferric reducing antioxidant power (FRAP)

The FRAP assay was performed following the method of Benzie and Strain with some modifications [13]. Briefly, 30 µl samples (1 g/l) or the standard were added to 270 µl of freshly prepared FRAP reagent and mixed thoroughly. The absorbance was measured at 595 nm after 30 min. Ferrous sulfate ($FeSO_4$) was used as the standard. The antioxidant potential of the sample was estimated against a standard curve of ferrous sulfate and the FRAP value was expressed as mg Fe(II) equivalents per gram of extract.

Results and discussion

Chemical composition

β-Glucan is a polysaccharide which is typically found in various species of mushrooms including reishi, shiitake and maitake [14]. β-Glucans have many functional and bioactive properties including anticancer abilities, antioxidant effects and activities against infectious diseases [14, 15]. The wild edible mushrooms studied were a rich source of fungal β-glucan with values ranging between 6.17 and 344.39 g/kg dw (Table 2). Shelf mushrooms were the richest sources of β-glucan followed by gilled mushrooms and puffball mushrooms. Park et al. [16] and Toledo et al. [17] found that the concentration of β-glucan in mushrooms varied from 76.0–101.0 and 27.0–89.2 g/kg dw, respectively. These values were considerably lower than results obtained in this study, with highest β-glucan concentration in *V. volvacea* (344.39 g/kg dw), followed by *L. squarrosulas* (191.85 g/kg dw), *R. violeipes* (136.61 g/kg dw) and *C. cibarius* (132.91 g/kg dw). Among the wild edible mushrooms studied, *A. trappei* had the lowest β-glucan content (6.17 g/kg dw) while β-glucan was not detected in *A. auricula*. *V. volvacea* and *L. squarrosulas* showed the greatest potential for commercial exploitation.

Mushrooms have high concentrations of carbohydrates and proteins. However, protein concentrations are affected by a number of factors including mushroom species, stage of development, the part sampled, level of nitrogen available, location, and the particular characteristics of the natural habitat [18]. This report studied the chemical composition of the twenty-five most popular wild edible mushroom species among the local Isan population. Chemical compositions (expressed on dry weight basis) for the selected mushroom species are shown in Table 2.

Table 3 Concentrations of individual phenolic compounds in twenty-five wild edible mushroom species located in Thailand

Sample	Flavonols					Flavan-3-ols			Flavanone	Total HPLC polyphenols
	Quercetin	Quercetin-3-*O*-rutinoside	Myricetin	Kaempferol	Total flavonols	(+)-catechin	(−)-epicatechin	Total flavan-3-ols	Naringenin	
Gilled mushrooms										
A. princeps	0.51 ± 0.02	0.62 ± 0.04	0.07 ± 0.02	0.03 ± 0.01	1.24 ± 0.11	3.01 ± 0.12	4.70 ± 0.24	7.71 ± 0.51	0.11 ± 0.00	9.06 ± 0.89
A. hamibapa	0.31 ± 0.01	0.30 ± 0.02	0.06 ± 0.00	0.01 ± 0.00	0.68 ± 0.05	2.67 ± 0.14	0.91 ± 0.14	3.58 ± 0.40	0.17 ± 0.03	4.43 ± 0.65
L. volemus	0.12 ± 0.01	0.24 ± 0.01	ND	ND	0.37 ± 0.03	1.91 ± 0.17	0.20 ± 0.03	2.10 ± 0.28	ND	2.47 ± 0.45
R. luteotacta	1.84 ± 0.08	0.70 ± 0.07	0.26 ± 0.03	0.01 ± 0.00	2.81 ± 0.23	3.11 ± 0.20	6.09 ± 0.42	9.20 ± 0.88	0.59 ± 0.02	12.61 ± 1.61
R. emetica	0.10 ± 0.01	0.14 ± 0.01	0.15 ± 0.01	0.02 ± 0.00	0.42 ± 0.04	1.70 ± 0.02	1.32 ± 0.21	3.03 ± 0.31	0.31 ± 0.04	3.75 ± 0.59
R. alboareolata	0.52 ± 0.02	0.13 ± 0.00	ND	ND	0.65 ± 0.03	0.41 ± 0.02	0.46 ± 0.04	0.86 ± 0.09	0.34 ± 0.03	1.85 ± 0.22
R. galochroides	0.42 ± 0.01	0.13 ± 0.02	ND	ND	0.54 ± 0.03	0.82 ± 0.04	0.55 ± 0.03	1.37 ± 0.11	ND	1.92 ± 0.20
R. cyanoxantha	0.43 ± 0.03	0.10 ± 0.01	ND	ND	0.53 ± 0.05	0.66 ± 0.08	0.66 ± 0.06	1.32 ± 0.20	ND	1.86 ± 0.37
R. nigricans	0.07 ± 0.01	0.26 ± 0.03	0.09 ± 0.02	0.02 ± 0.00	0.44 ± 0.04	1.55 ± 0.11	0.43 ± 0.03	1.99 ± 0.21	0.23 ± 0.01	2.66 ± 0.38
R. densfolia	0.22 ± 0.02	0.04 ± 0.00	0.17 ± 0.02	0.03 ± 0.00	0.45 ± 0.06	0.21 ± 0.02	1.01 ± 0.06	1.23 ± 0.12	0.59 ± 0.02	2.27 ± 0.30
R. delica	0.14 ± 0.01	0.26 ± 0.06	0.02 ± 0.00	0.04 ± 0.01	0.47 ± 0.08	2.83 ± 0.18	1.67 ± 0.09	4.50 ± 0.39	0.05 ± 0.00	5.02 ± 0.67
R. violeipes	0.09 ± 0.01	0.72 ± 0.04	0.09 ± 0.01	0.02 ± 0.00	0.93 ± 0.07	6.48 ± 0.34	2.04 ± 0.14	8.52 ± 0.69	0.21 ± 0.01	9.66 ± 1.09
T. fuliginosus	0.73 ± 0.02	0.12 ± 0.01	0.02 ± 0.00	0.01 ± 0.00	0.88 ± 0.03	8.43 ± 0.50	2.81 ± 0.10	11.24 ± 0.85	0.19 ± 0.01	12.31 ± 1.27
T. clypeatus	0.92 ± 0.04	0.27 ± 0.03	0.07 ± 0.02	ND	1.28 ± 0.11	13.40 ± 0.74	1.29 ± 0.24	14.69 ± 0.98	0.74 ± 0.06	16.71 ± 1.63
T. crassum	0.11 ± 0.01	0.14 ± 0.01	0.05 ± 0.00	ND	0.31 ± 0.04	2.09 ± 0.12	0.31 ± 0.05	2.40 ± 0.24	0.02 ± 0.00	2.73 ± 0.40
V. volvacea	1.83 ± 0.01	1.17 ± 0.02	0.68 ± 0.06	0.04 ± 0.01	3.72 ± 0.44	6.73 ± 0.69	5.87 ± 0.38	12.61 ± 0.52	0.72 ± 0.02	17.04 ± 1.40
Puffball mushrooms										
A. hygrometricus	0.04 ± 0.61	0.09 ± 0.04	0.04 ± 0.01	ND	0.18 ± 0.02	2.61 ± 0.11	3.78 ± 0.17	6.38 ± 0.39	0.28 ± 0.01	6.84 ± 0.61
A. trappei	ND	ND	ND	ND	ND	0.12 ± 0.01	0.20 ± 0.01	0.32 ± 0.01	ND	0.33 ± 0.02
Shelf mushrooms										
A. auricula	0.05 ± 0.50	0.08 ± 0.03	0.08 ± 0.01	ND	0.23 ± 0.02	1.52 ± 0.16	0.40 ± 0.04	1.91 ± 0.28	0.05 ± 0.13	2.19 ± 0.44
C. cibarius	0.24 ± 0.01	0.37 ± 0.02	0.03 ± 0.00	ND	0.88 ± 0.07	0.51 ± 0.04	0.39 ± 0.03	0.90 ± 0.10	0.54 ± 3.19	2.32 ± 0.30
Cra C. aureus	0.10 ± 0.00	0.38 ± 0.01	0.03 ± 0.00	ND	0.52 ± 0.03	0.68 ± 0.06	0.37 ± 0.06	1.05 ± 0.17	0.40 ± 3.07	1.97 ± 0.33
L. squarrosulas	0.10 ± 0.00	0.46 ± 0.03	0.17 ± 0.02	0.02 ± 0.00	0.76 ± 0.10	4.86 ± 0.16	1.57 ± 0.10	6.42 ± 0.37	0.09 ± 0.58	7.28 ± 0.68
L. polychrous	0.40 ± 0.02	0.77 ± 0.05	0.46 ± 0.05	0.06 ± 0.01	1.68 ± 0.23	5.61 ± 0.23	2.09 ± 0.12	7.70 ± 0.50	0.09 ± 1.13	9.48 ± 1.06
L. edodes	0.39 ± 0.01	0.49 ± 0.04	0.16 ± 0.01	0.02 ± 0.00	1.05 ± 0.09	3.05 ± 0.19	1.90 ± 0.16	4.96 ± 0.49	0.04 ± 0.62	6.05 ± 0.85
L. giganteus	ND	0.12 ± 0.00	0.02 ± 0.00	ND	0.14 ± 0.01	0.66 ± 0.06	0.98 ± 0.08	1.64 ± 0.20	ND	1.78 ± 0.30

Data were expressed g kg^{-1}dw (mean ± SD) (n = 10). ND, not detected

Total flavonols, qurercetin, quercetin-3-*O*-rutinoside, myricetin and kaempferol; total flavan-3-ols, (+)-catechin and (−)-epicatechin; total HPLC polyphenols, qurercetin, quercetin-3-*O*-rutinoside, myricetin, kaempferol, (+)-catechin, (−)-epicatechin, and naringenin

Table 4 Total phenolic concentration, total flavonoid concentration and antioxidant activity of twenty-five wild edible mushrooms species in Thailand

Sample	Total phenolics (g GAE kg^{-1} dw)[a]	Total flavonoids (g CE kg^{-1} dw)[b]	DPPH (% scavenging activity)	FRAP (g Fe(II) kg^{-1} dw)
Gilled mushrooms				
A. princeps	1.68 ± 0.38	0.62 ± 0.02	59.40 ± 1.19	3.01 ± 0.26
A. hamibapa	8.53 ± 0.91	0.81 ± 0.03	71.75 ± 8.94	7.54 ± 0.14
L. volemus	3.61 ± 0.36	0.52 ± 0.01	66.75 ± 2.18	0.92 ± 0.08
R. luteotacta	4.63 ± 0.20	2.09 ± 0.05	81.24 ± 5.10	7.53 ± 0.45
R. emetica	1.71 ± 0.46	0.75 ± 0.03	46.31 ± 2.16	0.20 ± 0.04
R. alboareolata	4.68 ± 0.61	1.06 ± 0.02	62.71 ± 1.35	2.66 ± 0.09
R. galochroides	2.38 ± 0.35	1.38 ± 0.01	69.76 ± 0.94	3.86 ± 0.23
R. cyanoxantha	2.91 ± 0.48	1.64 ± 0.02	78.74 ± 1.14	3.03 ± 0.36
R. nigricans	2.31 ± 0.26	1.03 ± 0.03	51.90 ± 2.06	0.32 ± 0.01
R. densfolia	5.62 ± 0.42	1.47 ± 0.05	55.43 ± 1.17	2.14 ± 0.02
R. delica	4.83 ± 0.42	1.75 ± 0.01	75.72 ± 1.18	3.74 ± 0.47
R. violeipes	2.26 ± 0.31	1.37 ± 0.06	63.59 ± 1.59	3.34 ± 0.11
T. fuliginosus	6.31 ± 0.50	2.19 ± 0.08	72.34 ± 8.11	4.47 ± 0.51
T. clypeatus	8.84 ± 0.53	5.10 ± 0.02	83.07 ± 2.41	9.79 ± 0.38
T. crassum	2.56 ± 0.54	1.53 ± 0.01	64.18 ± 1.77	0.37 ± 0.06
V. volvacea	8.49 ± 0.44	3.45 ± 0.08	86.60 ± 1.25	8.42 ± 0.67
Puffball mushrooms				
A. hygrometricus	2.33 ± 0.68	1.49 ± 0.03	45.72 ± 1.18	0.41 ± 0.04
A. trappei	0.72 ± 0.02	0.78 ± 0.03	36.75 ± 3.74	0.24 ± 0.04
Shelf mushrooms				
A. auricula	0.95 ± 0.25	0.15 ± 0.00	40.94 ± 0.73	0.10 ± 0.02
C. cibarius	1.41 ± 0.16	0.27 ± 0.01	64.10 ± 5.20	1.94 ± 0.39
Cra C. aureus	2.34 ± 0.55	0.21 ± 0.03	59.91 ± 1.35	4.12 ± 0.62
L. squarrosulas	5.42 ± 0.49	1.19 ± 0.02	71.68 ± 0.52	2.64 ± 0.18
L. polychrous	5.35 ± 0.14	2.27 ± 0.04	85.43 ± 1.04	3.91 ± 0.27
L. edodes	2.21 ± 0.46	0.24 ± 0.01	63.44 ± 3.64	2.50 ± 0.64
L. giganteus	1.46 ± 0.15	0.21 ± 0.00	56.90 ± 7.07	3.65 ± 0.50

[a]Total phenolic content (TPC) quantified by the Folin-Ciocalteu assay [g gallic acid equivalents (GAE) kg^{-1} dw]

[b]Total flavonoid content (TFC) quantified by the $AlCl_3$ method [g catechin equivalent (CE) kg^{-1} dw]. Results are expressed as mean values ± SD, $n = 10$. Antioxidant capacity measured by DPPH (% scavenging activity) and FRAP [g ferrous ion equivalent (g Fe kg^{-1} dw] assay

Protein was present at high concentrations and varied between 73.42 g/kg dw (*A. auricula*) and 487.10 g/kg dw (*T. crassum*). *C. aureus*, *A. hygrometricus*, *R. nigricans*, and *R. alboareolata* all recorded high protein contents at 223.84, 264.81, 285.51 and 320.84 g/kg dw, respectively; twice the values reported by Sanmee et al. [2]. Differences in protein concentration may result from location and ecological conditions. In general, wild mushrooms are richer sources of protein than commercial mushrooms. The protein content of sixteen gilled fungi (222.30–487.10 g/kg dw) was higher than that reported in oyster mushrooms (177.01–235.00 g/kg dw) [19].

L. edodes had the highest contents of sucrose and fructose (155.61 and 57.11 g/kg dw, respectively), while the highest glucose content was found in *R. nigricans* (23.86 g/kg dw) (Table 2). Fructose was present in all mushroom species except in *A. auricula* where it was not detected. Fructose and glucose concentration of *R. alboareolata* and *A. hygrometricus* were 17–20 times higher than those reported by Sanmee et al. [2]. However, here, glucose concentrations (0.07–23.90 g/kg dw) were similar to Croatian wild edible mushrooms (2.80–15.20 g/kg dw) [20] and Korean edible mushrooms (1.50–36.30 g/kg dw) [21].

The sugar alcohols mannitol, arabitol, xylitol and sorbitol were all found in the wild edible mushrooms studied (Table 2). Myo-inositol was only detected in *R. galochroides*, *R. luteotacta*, *L. polychrous* and *R. violeipes*. Mannitol, one of the phytochemicals of edible mushrooms, has the capacity to reduce blood pressure [22]. It also has diuretic effects [23]. With the exception of *A. trappei* and *A. auricula*, mannitol was present as the main polyol in all the wild edible mushrooms studied, with values ranging from 0.87 to 164.58 g/kg dw. Highest concentrations of mannitol were found in *L. volemus* (164.58 g/kg dw), *R. galochroides* (132.87 g/kg dw), *R. cyanoxantha* (131.42 g/kg dw), and *R. luteotacta* (126.13 g/kg dw), while the lowest was *C. cibarius* at 0.87 g/kg dw. Gilled mushrooms provided have the richest sources of mannitol. Arabitol was predominant in *R. galochroides, R. luteotacta*, *R. delica* and *R. nigricans* while the highest amount of sorbitol was detected in *T. crassum*. Arabitol has not previously been measured in *A. hygrometricus* [2] but was detected in this study.

Phenolic compounds profile

Limited investigations have been carried out regarding the individual profiles of phenolic compounds in wild edible mushrooms. Cultivated mushroom species are better known in terms of content; however, wild edible mushrooms are scarcely investigated and, to the best of our knowledge, the phenolic content of these twenty-five species has not been previously described. The phenolics, (+)-catechin and (−)-epicatechin were present in all twenty-five mushrooms (Table 3). Other phenolic compounds such as quercetin, quercetin-3-*O*-rutinoside, myricetin, naringenin and kaempferol were also detected. Quercetin and quercetin-3-*O*-rutinoside were the major flavonols ranging from 0.04–1.84 and 0.04–1.17 g/kg dw, respectively. The highest concentrations of quercetin were seen in *R. luteotacta* (1.84 g/kg dw) and *V. volvacea* (1.83 g/kg dw) while *V. volvacea* also had the highest concentrations of quercetin-3-*O*-rutinoside (1.17 g/kg dw) and myricetin (0.68 g/kg dw). Myricetin was detected in low concentrations in *R. delica*, *T. fuliginosus*, *C. aureus*, and *L. giganteus* ranging from 0.02 to 0.03 g/kg dw. However, these results were within the range of eight other edible mushrooms, *A. bisporus*, *B. edulis*, *C. gambosa*, *C. cibarius*, *C. cornucopioides*, *H. marzuolus*, *L. deliciosus* and *P. ostreatus* where values varied from 0.02 to 0.04 g/kg dw [24]. The most abundant flavonoids were (+)-catechin and (−)-epicatechin (flavan-3-ols) ranging from 0.12 to 13.40 g/kg dw and 0.20 to 6.09 g/kg dw, respectively. Total concentrations of the flavan-3-ols varied from 0.32 to 14.69 g/kg dw with the lowest concentration occurring in *A. trappei* and the highest in *T. clypeatus*. Naringenin is a flavanone that is also present in wild edible mushrooms, although less abundant than other flavonoids. It was the main component of the flavanone groups in the studied mushrooms (0.02–0.74 g/kg dw). The studied edible mushrooms had low quantities of kaempferol which occurs in some mushroom species in similar concentration (varying from 0.01 to 0.04 g/kg dw), except in *L. polychrous* where the kaempferol content was considerably higher (0.06 g/kg dw) than other mushrooms.

Total phenolic and flavonoid contents

As the antioxidation properties of extracts were often linked to their concentrations of phenolic and flavonoid compounds, the mushroom extracts were tested for their TPC and TFC. The amount of TPC was calculated as gallic acid equivalents, while catechin was used as the reference for the TFC assay. *T. clypeatus* had the highest TPC (8.84 g GAE/kg dw) among the wild edible mushroom species evaluated (Table 4), followed by *A. hamibapa* and *V. volvacea* with values of 8.53 and 8.49 g GAE/kg dw, respectively. The TFC in *T. clypeatus* was the highest (5.10 g CE/kg dw), while the lowest TFC was exhibited in *C. aureus* (0.21 g CE/kg dw). These results were lower than those reported in Portuguese wild edible mushrooms which varied from 4.58–58.14 g GAE/kg dw and 1.78–33.00 g CE/kg dw for TPC and TFC, respectively [25]. However, they were within the range of edible mushrooms in Malaysia where *H. tessulatus*, *P. eryngii*, *P.*

florida, *A. polytricha* and *F. velutipes* varied from 0.90–6.03 g GAE/kg dw and 0.20–6.95 g QE/kg dw for TPC and TFC, respectively [26]. The American Cancer Society has established 0.10 g per day of flavonoids as adequate for the mitigation of cancer and deteriorating illnesses [27]. Results for the wild edible mushrooms studied revealed that they all contained high flavonoids. The TFC values for *T. clypeatus, V. volvacea* and *L. polychrous* were 5.10, 3.45 and 2.27 g CE/kg dw, respectively, above the value suggested for daily intake by the American Cancer Society.

DPPH radical-scavenging activity

Oxidative stress is considered as being one of the most serious causes of various diseases; therefore, searching for natural scavengers of reactive oxygen species is highly desirable. Phenolic compounds may contribute directly to antioxidation activity [28] and it has been reported that the antioxidant activity of mushroom extracts correlated well with the content of their phenolic compounds [25]. Highly effective free radical scavenging activity of more than 80% was seen in the extracts of *V. volvacea* (86.6%), *L. polychrous* (85.4%), *T. clypeatus* (83.1%), and *R. luteotacta* (81.2%) (Table 4). These results were lower than those reported in Tanzania for wild edible mushrooms with values of 94.1% and 93.5% for *Cantharellus congolensis* and *Cantharellus cyanoxanthus*, respectively [29]. However, the methanolic extract of *L. volemus* showed stronger scavenging activities than those reported by Keleş et al. [30]. Khatua et al. [31] mentioned that the ethanolic extract from *R. delica* scavenged DPPH radicals by 60% at 1.4 mg/mL, lower than values reported here. The least activity was observed in puffball mushrooms *A. hygrometricus* (45.7%) and *A. trappei* (36.8%). These results revealed that mushroom extracts with higher phenolic compounds also had higher scavenging activity. This finding has also been observed in other studies including Palacios et al. [24], Pereira et al. [25] and Tibuhwa [29]. Here, *V. volvacea* was observed to have high TPC and TFC values of 8.49 g GAE/kg dw and 3.45 g CE/kg dw, respectively, corresponding to a high scavenging capacity of 86.6%. In contrast, *A. trappei* had the lowest TPC and TFC values (0.72 g GAE/kg dw and 0.78 g CE/kg dw, respectively) and exhibited the lowest scavenging ability (36.8%).

Ferric reducing antioxidant power (FRAP)

Antioxidant capacity was estimated by the FRAP method, which determines the capacity of antioxidant components to reduce a Fe^{3+}-TPTZ complex to Fe^{2+}-TPTZ. Hence, a higher Fe^{3+}-TPTZ reduction means a higher antioxidant capacity. The reducing power of antioxidant components may serve as an indicator of potential antioxidant capacity [32]. Reducing powers of *T. clypeatus*, *V. volvacea*, *A. hamibapa* and *R. luteotacta* were very high (9.79, 8.42, 7.54 and 7.53 g Fe(II)/kg dw, respectively). Reducing powers of *T. clypeatus* and *V. volvacea* were higher than those of the other mushrooms as they contained higher TPC and TFC (Table 4). Reducing powers of gilled mushrooms were generally higher than shelf mushrooms and puffball mushrooms. High antioxidant capacities in mushrooms can suppress reactive oxygen species which are related to aging and deteriorating illnesses [33]. They might also serve as a good resource for the future development of functional foods or drugs.

This is the first research into the phenolic composition and antioxidant activity of native wild edible mushroom species in Thailand. The study focused particularly on the antioxidant potential of twenty-five wild edible mushroom species from Northeastern Thailand and determined their chemical composition in terms of proteins, sugars, β-glucan, and phenolic compounds. Phenolic compounds were the main components responsible for the antioxidant capacity of all the wild mushrooms studied. *T. clypeatus* and *V. volvacea* demonstrated the strongest reducing powers and radical scavenging activities, as well as having the highest concentrations of phenolic compounds. Additionally, *T. clypeatus* and *V. volvacea* exhibited the highest amounts of β-glucan and protein. These findings suggest *T. clypeatus* and *V. volvacea* as highly recommended for consumption. Further research is required to establish the intracellular antioxidant activity of wild edible mushroom extracts prior to application as food supplements or in the development of nutraceuticals.

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References

1. Puttaraju NG, Venkateshaiah SU, Dharmesh SM, Nanjaraj Urs SM, Somasundaram R. Antioxidant activity of indigenous edible mushrooms. J. Agric. Food Chem. 54: 9764–9772 (2006)
2. Sanmee R, Dell B, Lumyong P, Izumori K, Lumyong S. Nutritive value of popular wild edible mushrooms from northern Thailand. Food Chem. 82: 527–532 (2003)
3. Barros L, Cruz T, Baptista P, Estevinho L, Ferreira I. Wild and commercial mushrooms as source of nutrients and nutraceuticals. Food Chem. Toxicol. 46: 2742–2747 (2008)
4. Wasser SP. Medicinal mushrooms as a source of antitumor and immunomodulating polysaccharides. Appl. Microbiol. Biotechnol. 60: 258–274 (2002)
5. Ada MA, Wong HK, Jiang B. Fibrinolytic enzymes in Asian traditional fermented foods. Food Res. Int. 38: 243–250 (2005)

6. Barros L, Calhelha RC, Vaz JA, Ferreira ICFR, Baptista P, Estevinho LM. Antimicrobial activity and bioactive compounds of Portuguese wild edible mushrooms methanolic extracts. Eur. Food Res. Technol. 225: 151–156 (2007)
7. Mau JL, Tsai SY, Tseng YH, Huang SJ. Antioxidant properties of hot water extracts from *Ganoderma tsugae* Murrill. LWT-Food Sci. Technol. 38: 589–597 (2005)
8. Seephonkai P, Samchai S, Thongsom A, Sunaart S, Kiemsanmuang B, Chakuton K. DPPH radical scavenging activity and total phenolics of *Phellinus* mushroom extracts collected from Northeast of Thailand. Chin. J. Nat. Med. 9: 441–445 (2011)
9. Kelebek H, Selli S, Canbas A, Cabaroglu T. HPLC determination of organic acids, sugars, phenolic compositions and antioxidant capacity of orange juice and orange wine made from a Turkish cv. Kozan. Microchem. J. 91: 187–192 (2009)
10. Singleton VL, Rossi JA. Colorimetry of total phenolics with phosphomolybdic- phosphotungstic acid reagents. Am. J. Enol. Viticult. 16: 144–158 (1965)
11. Kim D, Chun O, Kim Y, Moon H, Lee C. Quantification of phenolics and their antioxidant capacity in fresh plums. J. Agric. Food Chem. 51: 6509–6515 (2003)
12. Brand-Williams W, Cuvelier ME, Berset C. Use of a free radical method to evaluate antioxidant activity. LWT-Food Sci. Technol. 28: 25–30 (1995)
13. Benzie IFF, Strain JJ. Ferric reducing/antioxidant power assay: Direct measure of total antioxidant activity of biological fluids and modified version for simultaneous measurement of total antioxidant power and ascorbic acid concentration. Methods Enzymol. 299: 15–27 (1999)
14. Wasser SP, Weis AL. Therapeutic effects of substances occurring in higher basidiomycetes mushrooms: a modern perspective. Crit. Rev. Immunol. 19: 65–96 (1999)
15. Brown GD, Gordon S. Fungal β-glucans and mammalian immunity. Immunity 19: 311–315 (2003)
16. Park YK, Ikegaki M, Alencar SM, Aguiar AL. Determinação da concentração de β-glucano em cogumelo *Agaricus blazei* Murill por método enzimático. Cienc. y Tecnol. Aliment. 23: 312–316 (2003)
17. Toledo RCC, Carvalho MA, Lima LCO, Vilas-Boas EVB, Dias ES. Measurement of β-glucan and other nutritional characteristics in distinct strains of *Agaricus subrufescens* mushrooms. Afr. J. Biotechnol. 12: 6203–6209 (2013)
18. Flegg PB, Maw G. Mushrooms and their possible contribution to world protein needs. Mushroom J. 48: 395–403 (1977)
19. Khan A, Amin SMR, Uddin N, Tania M, Alam N. Comparative study of the nutritional composition of oyster mushrooms cultivated in Bangladesh. Bangladesh J. Mushroom. 2: 9–14 (2008)
20. Beluhan S, Ranogajec A. Chemical composition and non-volatile components of Croatian wild edible mushrooms. Food Chem. 124: 1076–1082 (2011)
21. Kim M-Y, Chung L-M, Lee S-J, Ahn J-K, Kim E-H, Kim M-J, Kim S-L, Moon H-I, Ro H-M, Kang E-Y, Seo S-H, Song H-K. Comparison of free amino acid, carbohydrates concentrations in Korean edible and medicinal mushrooms. Food Chem. 113: 386–393 (2009)
22. Hagiwara S, Takahashi M, Shen Y, Kaihou S, Tomiyama T, Yazawa M, Tamal Y, Sin Y, Kazusaka A, Terazawa M. A phytochemical in edible Tamogi-take mushroom (*Pleurotus cornucopiae*), D-mannitol, Inhibit ACE activity and lowers the blood pressure of spontaneously hypertensive rats. Biosci. Biotechnol. Biochem. 69: 1603–1605 (2005)
23. Zhao Y, Xie R, Chao X, Zhang Y, Lin R, Sun W. Bioactivity-directed isolation, identification of diuretic compounds from *Polyporus umbellatus*. J. Ethnopharmacol. 126: 184–187 (2009)
24. Palacios I, Lozano M, Moro C, D'Arrigo M, Rostagno MA, Martínez JA, García- Lafuente A, Guillamón E, Villares A. Antioxidant properties of phenolic compounds occurring in edible mushrooms. Food Chem. 128: 674–678 (2011)
25. Pereira E, Barros L, Martins A, Ferreira ICFR. Towards chemical and nutritional inventory of Portuguese wild edible mushrooms in different habitats. Food Chem. 130: 394–403 (2012)
26. Wong F-C, Chai T-T, Tan S-L, Yong, A-L. Evaluation of bioactivities and phenolic content of selected edible mushrooms in Malaysia. Trop. J. Pharm. Res. 12: 1011–1016 (2013)
27. Krebs-Smith SM, Cook DA, Subar AF, Cleve-land L, Friday J. US Adults' fruit and vegetable intakes, 1989 to 1991: A revised baseline for the healthy people 2000 objective. Am. J. Public Health. 85: 1623–1629 (2000)
28. Duh PD, Tu YY, Yen GC. Antioxidant activity of water extract of harn jyur (*Chyrsanthemum morifolium* Ramat). Lebensm. Wiss. Technol. 32: 269–277 (1999)
29. Tibuhwa DD. A comparative study of antioxidant activities between fresh and dry mushrooms in the genera *Cantharellus* and *Afrocantharellus* from Tanzania. Food Nutr. Sci. 5: 212–221 (2014)
30. Keleş A, Koca I, Gençcelep H. Antioxidant properties of wild edible mushrooms. J. Food Process. Technol. 2: 1–6 (2011)
31. Khatua S, Paul S, Chatterjee A, Ray D, Roy A, Acharya K. Evaluation of antioxidative activity of ethanolic extract from *Russula delica*: an in vitro study. J. Chem. Pharm. Res. 5: 100–107 (2013)
32. Kalogeropoulos N, Yanni AE, Koutrotsios G, Aloupi M. Bioactive microconstituents and antioxidant properties of wild edible mushrooms from the island of Lesvos, Greece. Food Chem. Toxicol. 55: 378–385 (2013)
33. Yang JH, Lin HC, Mau JL. Antioxidant properties of several commercial mushrooms. Food Chem. 77: 229–235 (2002)